

# FR-016 M0 — Multi-Agent Orchestration Engineering Spike

FR016 — `sources/optional/m0_spike.md`

Masterclass source package snapshot

|                        |                                                                    |
|------------------------|--------------------------------------------------------------------|
| Functional Requirement | FR-016                                                             |
| Edition                | Engineering Learning Academy — Source Package snapshot             |
| Status                 | <b>**Accepted with revisions**</b> (owner 2026-08-06)              |
| Audience               | Academy generation and study (snapshot; SoT remains authoritative) |
| Document type          | Source package snapshot (PDF)                                      |

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via `python scripts/build_masterclass_package.py`. Repository documentation remains the source of truth.

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# FR-016 M0 — Multi-Agent Orchestration Engineering Spike

**Status:** Accepted with revisions (owner 2026-08-06)

**Date:** 2026-08-06

**Phase:** Horizon 1A Stage 10

**ADR:** [ADR-008](#) (Accepted at M1)

**Scope (M0 — historical):** Architecture and planning only. No production implementation in this milestone.

**Preceding capability:** [FR-015 Bounded Agentic Workflow](#) (complete and frozen); [ADR-007](#).

**Operational evidence:** [OAT-001 Phase 3](#), [Phase 4 polish](#), [reconciliation](#).

**Succeeded by:** [M1 contracts](#); FR-016 later **Complete / Frozen** — [acceptance](#)

**Does not begin (M0):** production runtime, job discovery, auto-submission, FR-017 implementation, or study-aid generation.

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## 1. Executive Summary

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FR-016 asks whether CIC needs **more than one bounded agent**, and if so what smallest safe orchestration architecture adds genuine value over FR-008 deterministic workflow and FR-015 single-agent coordination (BOPA).

**Critical finding:** Splitting BOPA into Prep / Truth / Review “specialists” that wrap the same services with the same mutating tools is **multi-agent theatre**. It fails the dual-value test, weakens explainability, and teaches the wrong interview lesson.

**Second finding:** After OAT-001, BOPA is **operationally ready** for single-Opportunity post-acquisition coordination. Remaining friction was presentation and historical corpus hygiene — not an authority or topology gap that requires a specialist team.

**Third finding:** A genuine multi-agent need appears when **permission boundaries**, **failure domains**, or **future acquisition hand-in** diverge. Today that second live boundary barely exists inside Horizon 1A scope (job discovery and autonomous submission remain forbidden). The learning and substrate case for FR-016 is stronger than the near-term commercial case.

**Recommended architecture (if FR-016 proceeds now):**

**Deterministic Orchestration Supervisor (DOS) + typed specialist runtimes with per-specialist ToolPolicy + append-only typed handoffs.** Start with two live specialists only:

1. **BOPA** — existing preparation/truth coordination (mutating allow-list unchanged).
2. **Operational Briefing Specialist (OBS)** — read-only cross-surface diagnosis and owner briefing; **no** preparation, truth mutation, submit, or pipeline tools.

Supervisor **delegates only**. It never inherits specialist authority, never waives truth, never submits, never mutates pipeline, and never re-enters FR-008.

**Decision policy: Deterministic supervisor routing by typed orchestration state** is normative. Optional LLM may propose a specialist or owner-facing summary under the same deterministic DelegationPolicy — never as authority. Deterministic specialist proposers remain the operational default (FR-015 M4 / OAT-001).

**Framework:** Do **not** adopt LangGraph, OpenAI Agents SDK, Semantic Kernel / Microsoft Agent Framework, or CrewAI in FR-016. Extend the custom `career_intelligence.agent` contracts. Keep architecture portable so frameworks remain teaching analogies, not dependencies.

**Commercial honesty:** If the owner's sole priority is faster applications this week, **defer FR-016** and keep using `cic agent`. If the owner accepts FR-016 as **bounded multi-agent substrate + interview-transferable skill** with modest near-term product upside, **accept the constrained architecture below**.

**M0 recommendation: Accept with conditions** — or **Defer** if commercial priority wins. Do not accept a theatrical specialist cast. Do not begin M1 until the owner chooses Accept / Accept-with-revisions / Defer.

## 2. Current Architecture and Bootstrap Findings

### 2.1 Layers already authoritative

| Layer                                                          | Owner                                                     | Agent implication                                                         |
|----------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|
| Acquire → analyse → assess → match → strategy → owner decision | FR-008 / <code>orchestration</code> / ADR-003             | Do not rebuild; do not re-enter from agents to "repair" missing artefacts |
| Opportunity SoT; review queue projection; duplicates linked    | FR-009 / ADR-004                                          | Agents cite Opportunity; never invent identity                            |
| Package composition                                            | FR-010                                                    | Thin adapter only                                                         |
| Preparation sequencing                                         | FR-011                                                    | Thin adapter only                                                         |
| Submission assistance + attempts                               | FR-012                                                    | Out of specialist write path                                              |
| Pipeline lifecycle + PipelineEvents                            | FR-013 / ADR-005                                          | Agent status ≠ pipeline status                                            |
| Truth validation + external-use gate                           | FR-014 / ADR-006                                          | Consume; never waive / rewrite / stale-pass                               |
| Bounded single-agent coordination                              | FR-015 / ADR-007 / <code>career_intelligence.agent</code> | Substrate to extend, not replace                                          |

## 2.2 BOPA runtime (inspected)

Live loop in `AgentRuntime`:

```
observe readiness → classify state class → proposer suggests AgentAction
→ evaluate_action_policy (sole admission) → thin adapter → audit → re-observe / stop
```

Allow-list (frozen): `inspect_readiness`, `run_preparation`, `verify_package`, `validate_truth_package`, `request_owner_review`, `stop`.

Forbidden (explicit): submit, pipeline mutation, discovery, recruiter contact, truth waive, Markdown rewrite, FR-008 analyse/assess/match/strategy tools, filesystem/shell/arbitrary code.

Audit: append-only `AgentRun` / events under `data/agent_runs/`. Domain SoTs remain elsewhere.

DeterministicActionProposer is the operational default; `--llm` optional under the same ToolPolicy ([fr015\\_m4\\_evaluation.md](#)).

## 2.3 OAT-001 lessons that bind FR-016

| Lesson                                                                   | FR-016 implication                                                             |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| BOPA safe on live corpus; no pipeline mutation; truth fail-closed        | Preserve ToolPolicy discipline per specialist                                  |
| Resume did not duplicate prep/truth                                      | Orchestration resume must re-inspect SoT + completed ops                       |
| Material-benefit / presentation issues fixed in Phase 4 without redesign | Do not invent specialists to paper over UX                                     |
| Deterministic default appropriate                                        | Do not make LLM supervisor normative                                           |
| Corpus identity / reconciliation mattered before realistic agent use     | Operational Briefing may cite identity/pipeline context but must not "fix" SoT |
| Dogfooding > automated tests alone                                       | Manual journeys remain mandatory                                               |

## 2.4 FR-015 foreshadowing for FR-016

FR-015 intentionally deferred agent-to-agent messaging, supervisor graphs, and shared blackboards. Contracts designed for reuse: `AgentAction`, `AgentRun`, `ToolPolicy`, `AgentStopReason`, audit events. That substrate is sufficient to extend; it does not obligate multi-agent product scope.

## 2.5 Generic multi-agent concepts vs CIC decisions

| Generic concept                 | CIC-specific decision                                                                |
|---------------------------------|--------------------------------------------------------------------------------------|
| Supervisor / workers            | Deterministic supervisor; specialists are bounded runtimes, not chat personas        |
| Handoff                         | Typed append-only record + DelegationPolicy; not free-form chat                      |
| Shared memory                   | Forbidden as SoT; re-read Opportunity/package/truth/pipeline/agent audit             |
| Agents-as-tools                 | Closest portable analogy: supervisor invokes specialist runs as bounded capabilities |
| Peer handoff ownership transfer | Rejected as default — supervisor retains orchestration ownership                     |
| Swarm / debate                  | Rejected                                                                             |

## 3. Problem Statement

The owner still faces coordination questions **across** agent runs, opportunities, and future capabilities:

1. When should preparation coordination run vs when should the system only brief?
2. How does a second specialist with a **different** allow-list get invoked without privilege escalation?
3. How are parent/child runs, handoffs, and global step/cost limits audited for FR-017?
4. How can a future Job Discovery capability hand an acquired Opportunity into bounded coordination without redesign?

FR-016's engineering question is not "can we stage a team of AI employees?" It is:

What problem requires more than one bounded agent, and what is the smallest orchestration that solves it without duplicating FR-008/FR-015, weakening FR-014, or adding LLM tax for sequencing that is already known?

If the answer collapses to "rename BOPA actions as agents," FR-016 should be deferred.

## 4. Evidence: Is Multi-Agent Justified — or Premature?

### 4.1 What FR-015 already solves

For one already-acquired Opportunity with `decision=apply`:

- Observe package / truth / approvals
- Run preparation / verify / validate truth when legal

- Stop for owner review / truth block / material benefit
- Resume without duplicate work
- Fail closed on injection and forbidden actions

OAT-001 confirmed this path is safe and, after Phase 4 polish, operationally ready.

## 4.2 What one BOPA cannot do (genuine gaps)

| Gap                                                           | Nature                        | Multi-agent needed?                                   |
|---------------------------------------------------------------|-------------------------------|-------------------------------------------------------|
| Different mutating vs read-only authority in one run tree     | Permission isolation          | <b>Yes</b> (if second specialist exists)              |
| Parent orchestration audit spanning multiple specialist runs  | Observability substrate       | <b>Yes</b> (orchestration layer; can be thin)         |
| Batch "brief me across N Opportunities" without mutating each | Parallel read-only            | <b>Maybe</b> (OBS + supervisor; not Prep/Truth split) |
| Job Discovery → acquired Opportunity → coordination           | Future hand-in                | <b>Later</b> (placeholder only)                       |
| Prep then truth as separate "agents"                          | Sequencing already in BOPA    | <b>No</b>                                             |
| Debate which claim is true                                    | Truth is deterministic FR-014 | <b>No</b>                                             |
| Auto-submit / pipeline advance                                | Forbidden                     | <b>No</b>                                             |

## 4.3 Verdict

| Lens                                    | Verdict                                                      |
|-----------------------------------------|--------------------------------------------------------------|
| Near-term job-acquisition product value | <b>Weak</b> — BOPA already removes the main coordination tax |
| Safety / authority isolation learning   | <b>Strong</b> — per-specialist ToolPolicy + typed handoffs   |
| FR-017 preparedness                     | <b>Strong</b> — parent/child traces, delegation metrics      |
| Interview / transferable skill          | <b>Strong</b> if architecture is honest                      |
| Theatrical specialist cast              | <b>Reject</b>                                                |
| Full deferral until Job Discovery       | <b>Valid</b> under dual-value / job-acquisition-first        |

**Spike position:** FR-016 is **justified as a constrained orchestration substrate** if the owner accepts modest near-term product upside and high learning value. It is **not** justified as a role-playing team replacing BOPA. If commercial-only priority wins, **defer**.

## 5. Genuine Value Scenarios

### Value matrix

| Scenario                                                                   | Current owner effort                 | FR-008 / FR-015 behaviour                    | Proposed specialist                              | Why separate agent?                   | Added complexity    | Measurable benefit                                                       | Recommendation                                 |
|----------------------------------------------------------------------------|--------------------------------------|----------------------------------------------|--------------------------------------------------|---------------------------------------|---------------------|--------------------------------------------------------------------------|------------------------------------------------|
| Messy Opportunity: apply + missing package + truth history + pipeline note | Interpret several CLIs / show output | BOPA mutates when legal; show already briefs | OBS (read-only) then optional BOPA               | Different ToolPolicy (no mutate)      | Medium              | Fewer accidental prep attempts on interviewing/ submitted; clearer brief | <b>Include</b> (narrow)                        |
| Happy-path prep → truth → owner review                                     | Low after FR-015                     | BOPA alone                                   | None                                             | No                                    | High if split       | None                                                                     | <b>Reject</b> split                            |
| Truth blocked remediation briefing                                         | Owner edits Markdown; revalidate     | BOPA stops; show lists blockers              | OBS may summarise blockers from TruthReport refs | Isolation of read-only briefing       | Low–medium          | Slight UX                                                                | <b>Include lightly</b> (OBS) or keep BOPA show |
| Batch readiness across open apply Opportunities                            | Manual loop                          | One BOPA run each                            | Supervisor fans out OBS then selective BOPA      | Orchestration + isolation             | Medium              | Time saved on triage                                                     | <b>Include</b> (M3+)                           |
| Illegal submit / waive proposal                                            | N/A                                  | ToolPolicy deny                              | Same                                             | Policy already works                  | None for new agents | Safety regression risk if broadened                                      | <b>Reject</b> new authority                    |
| Peer agent debate on ranking                                               | N/A                                  | FR-009 deterministic rank                    | None                                             | Ranking must stay deterministic       | High                | Negative                                                                 | <b>Reject</b>                                  |
| Job Discovery hand-in                                                      | Future                               | Out of scope                                 | Acquisition placeholder                          | Future permission boundary            | Low now             | Compatibility                                                            | <b>Defer</b> (placeholder contract only)       |
| Operational identity repair                                                | Manual / OAT scripts                 | Outside agent                                | None as agent                                    | SoT repair is engineering, not agency | High risk           | Negative if automated                                                    | <b>Reject</b> as agent                         |



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## 6. Alternative Architectures

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| ID | Alternative                                                  | Summary                                                            |
|----|--------------------------------------------------------------|--------------------------------------------------------------------|
| A  | Keep BOPA only                                               | No FR-016 product; document deferral                               |
| B  | Deterministic supervisor + bounded specialists               | DOS selects specialist by typed state; specialists keep ToolPolicy |
| C  | LLM proposer + deterministic supervisor policy + specialists | LLM suggests specialist/goal; DelegationPolicy admits              |
| D  | Coordinator agent with direct specialist handoffs            | LLM coordinator owns conversation; specialists take over           |
| E  | Hierarchical multi-agent team                                | Supervisor → leads → specialists                                   |
| F  | Framework-based orchestration                                | LangGraph / Agents SDK / MAF / CrewAI host the graph               |
| G  | Custom typed orchestration extending AgentRuntime contracts  | B/C without framework; portable contracts                          |

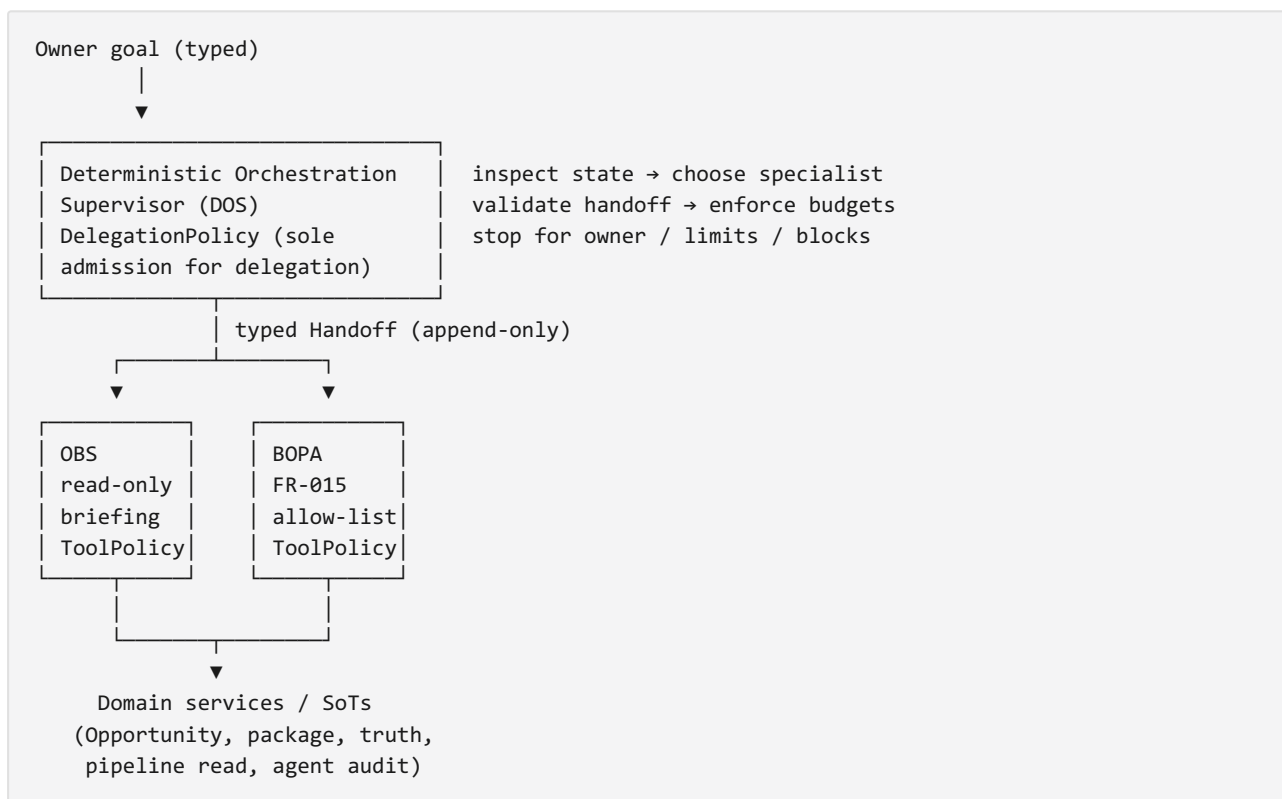
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## 7. Trade-off Analysis

| Criterion             | A Keep BOPA       | B Det. supervisor   | C LLM + det. policy           | D Coordinator handoffs      | E Hierarchy | F Framework          | G Custom typed  |
|-----------------------|-------------------|---------------------|-------------------------------|-----------------------------|-------------|----------------------|-----------------|
| Safety                | Highest (known)   | High                | High if policy sole authority | Medium (chat/control bleed) | Lower       | Depends on mapping   | High            |
| Explainability        | High              | Highest             | High                          | Medium                      | Low         | Medium               | Highest         |
| Deterministic control | High              | Highest             | High                          | Low–medium                  | Low         | Medium               | Highest         |
| Complexity            | Lowest            | Medium              | Medium+                       | High                        | Highest     | High+deps            | Medium          |
| Latency / tokens      | Lowest            | Low                 | Higher                        | Highest                     | Highest     | Variable             | Low             |
| Auditability          | High (single run) | High (parent/child) | High                          | Medium                      | Medium      | Tooling-dependent    | High            |
| Testability           | Proven            | Strong              | Strong w/ fixtures            | Weak                        | Weak        | Harder               | Strong          |
| Failure isolation     | Single domain     | Strong              | Strong                        | Weak                        | Weak        | Variable             | Strong          |
| Commercial value now  | Already delivered | Modest              | Modest                        | Low                         | Low         | Low                  | Modest          |
| Learning value        | Incremental       | High                | High                          | Medium (fashion risk)       | Low         | Medium (jargon risk) | High            |
| FR-008 conflict       | None              | None if separate    | None                          | Risk                        | Risk        | Risk                 | None if careful |

## 8. Recommended Topology

**Primary recommendation: B + G** — Deterministic Orchestration Supervisor over custom typed contracts; optional C as non-default proposal path.



**Rejected as default:** D (peer ownership transfer), E (hierarchy), F (framework now), Prep/Truth/Review persona cast.

#### Portable analogies (for interviews, not dependencies):

- OpenAI Agents SDK "agents as tools" ≈ supervisor retains control; specialist run returns typed result ([orchestration guide](#)).
- OpenAI "handoffs" ≈ ownership transfer — **not** CIC default.
- LangGraph supervisor/subagents ≈ same distinction; checkpoint only outer graph.
- Code-orchestrated agents (OpenAI docs) ≈ CIC deterministic supervisor.

## 9. Supervisor Responsibility and Authority

### Owns

- Inspect typed orchestration observation (derived; not SoT)
- Select an approved specialist for the current goal/state
- Create/validate typed handoffs via **DelegationPolicy**
- Enforce global step, visit, timeout, and cost budgets
- Aggregate specialist stop reasons into orchestration stop
- Persist orchestration audit / checkpoint
- Pause for owner gates; resume with SoT re-inspection

## Must not

- Bypass specialist ToolPolicy
- Call mutating domain services directly
- Approve external use / waive TruthReports
- Submit or advance pipeline
- Invent evidence or Opportunity fields
- Grant one specialist another's permissions via handoff
- Re-enter FR-008 graph to repair missing FR-002-005
- Become a second workflow engine

**Principle:** Supervisor authority is **delegation admission**, not domain super-user. Sitting above specialists does not expand tools.

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## 10. Candidate Specialist Agents

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### 10.1 Include

#### BOPA — Bounded Opportunity Preparation Agent (existing)

| Field                   | Definition                                                                                             |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Responsibility          | Coordinate prep/verify/truth toward owner review for one Opportunity                                   |
| Start                   | Owner/orchestration goal <code>prepare_for_owner_review</code> ; persisted <code>opportunity_id</code> |
| Stop                    | Existing <code>AgentStopReason</code> set                                                              |
| Allowed tools           | FR-015 allow-list                                                                                      |
| Forbidden               | FR-015 forbidden set                                                                                   |
| Input                   | <code>AgentGoal</code> + approvals flags                                                               |
| Output / handoff back   | Child <code>AgentRun</code> id, stop reason, artefact refs, snapshot hash                              |
| SoT                     | Opportunity / package / truth services                                                                 |
| Owner gate              | Document review, truth remediation, material-benefit override                                          |
| Failure modes           | Policy deny, adapter errors, provider down, max steps                                                  |
| Why not only a service? | Already justified in FR-015: state-aware next-action under policy                                      |

## OBS — Operational Briefing Specialist (new, narrow)

| Field                      | Definition                                                                                                                                                                                                                 |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Responsibility             | Read-only cross-surface diagnosis and owner-facing brief (pipeline note, truth blockers, whether BOPA is advisable)                                                                                                        |
| Start                      | Goal <code>brief_opportunity_readiness</code> or supervisor selects OBS when mutate is illegal/unwise                                                                                                                      |
| Stop                       | <code>briefing_complete</code> , <code>clarification_required</code> , <code>unsupported_state</code> , <code>budgets</code>                                                                                               |
| Allowed tools              | <code>inspect_readiness</code> , <code>inspect_pipeline_status</code> (read), <code>inspect_truth_blockers</code> (read), <code>inspect_agent_history</code> (read), <code>request_owner_review</code> , <code>stop</code> |
| Forbidden                  | All BOPA mutating tools; submit; pipeline write; truth waive; discovery; FR-008                                                                                                                                            |
| Input                      | <code>opportunity_id</code> , optional owner notes (never authority)                                                                                                                                                       |
| Output                     | Typed <code>OperationalBrief</code> (state class, recommended next specialist or owner action, evidence refs)                                                                                                              |
| SoT                        | Same domain reads; brief is derived                                                                                                                                                                                        |
| Owner gate                 | Always — brief never substitutes approval                                                                                                                                                                                  |
| Failure modes              | Missing Opportunity; contradictory state; budget                                                                                                                                                                           |
| Why not BOPA action alone? | Different ToolPolicy: cannot mutate even if snapshot would allow prep; enables safe batch triage and “brief first” orchestration without accidental preparation                                                            |

## 10.2 Reject

| Candidate                                                     | Why rejected                                                   |
|---------------------------------------------------------------|----------------------------------------------------------------|
| Application Preparation Specialist (wrap FR-011 only)         | Renames a service; BOPA already coordinates                    |
| Truth and External-Use Specialist (with mutate validate only) | Splitting prep/truth adds handoff cost; BOPA sequencing proven |
| Truth Agent that can waive / rewrite                          | Violates ADR-006                                               |
| Review/Handoff Specialist as chat persona                     | Owner review is human; no agent approves                       |
| Pipeline Agent with lifecycle authority                       | Violates ADR-005                                               |
| Submission Agent with autonomous submit                       | Violates FR-012 / human review                                 |
| Acquisition Specialist (live)                                 | Job discovery out of scope                                     |
| Ranking / prestige Agent                                      | Ranking deterministic; prestige bias forbidden                 |
| Debate / critic Agent pair                                    | Theatre; no dual-value                                         |

## 10.3 Placeholder only (contracts, no runtime)

**Future Acquisition Specialist** — accepts discovery envelopes **after** an Opportunity exists via FR-008/FR-009 path; may only hand `opportunity_id` into DOS. No Seek/LinkedIn/Indeed/scraping in FR-016.

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## 11. Agent / Tool Authority Matrix

| Capability                                   | DOS           | OBS             | BOPA                 | Owner              |
|----------------------------------------------|---------------|-----------------|----------------------|--------------------|
| Inspect derived orchestration state          | ✓             | ✓               | ✓                    | ✓                  |
| Delegate to OBS                              | ✓<br>(policy) | —               | —                    | via CLI            |
| Delegate to BOPA                             | ✓<br>(policy) | propose<br>only | —                    | via CLI            |
| run_preparation / verify / validate<br>truth | —             | —               | ✓                    | CLI services       |
| Read pipeline status                         | ✓             | ✓               | ✓<br>(informational) | ✓                  |
| Advance pipeline                             | —             | —               | —                    | ✓ only             |
| Submit                                       | —             | —               | —                    | ✓ only             |
| Waive truth                                  | —             | —               | —                    | never<br>automated |
| Grant cross-specialist tools                 | —             | —               | —                    | never              |

### Policy model recommendation

**Global orchestration policy + per-specialist ToolPolicy** (option 3).

1. **DelegationPolicy** — who may be invoked, from which states, with which goals.
2. **Specialist ToolPolicy** — unchanged pattern from FR-015; per allow-list.
3. Handoff cannot widen tools. Target policy re-evaluates on fresh snapshot.
4. Capability tokens optional later; not required if typed agent\_id + allow-list matrices are enforced in code.

Privilege escalation must be **impossible or fail-closed**.

## 12. Delegation and Handoff Contracts

### Handoff record (authoritative; append-only)

Proposed fields:

| Field                              | Purpose                                                                                     |
|------------------------------------|---------------------------------------------------------------------------------------------|
| <code>handoff_id</code>            | Unique id                                                                                   |
| <code>orchestration_run_id</code>  | Parent run                                                                                  |
| <code>source</code>                | <code>supervisor</code> or specialist id                                                    |
| <code>target_agent</code>          | <code>bopa</code>   <code>obs</code>   ...                                                  |
| <code>opportunity_id</code>        | Required                                                                                    |
| <code>requested_goal_kind</code>   | Typed goal                                                                                  |
| <code>observed_state_hash</code>   | Snapshot / orchestration observation hash                                                   |
| <code>input_artefact_refs</code>   | Package/truth/run refs — citations only                                                     |
| <code>preconditions</code>         | Structured flags                                                                            |
| <code>expected_output_kind</code>  | e.g. <code>agent_run_result</code> , <code>operational_brief</code>                         |
| <code>owner_approval_status</code> | Present / missing                                                                           |
| <code>policy_decision</code>       | Allow/deny + reason                                                                         |
| <code>reason</code>                | Short deterministic rationale                                                               |
| <code>created_at</code>            | Timestamp                                                                                   |
| <code>acceptance</code>            | <code>accepted</code>   <code>rejected</code>   <code>stale</code>   <code>cancelled</code> |

## Rules

- **Who may request:** Supervisor always; specialists may *recommend* next specialist only via typed output — supervisor (or owner CLI) must admit.
- **Who validates:** DelegationPolicy before create; target ToolPolicy before act.
- **Target reject:** Yes — if preconditions/hash fail on acceptance.
- **Stale state:** If SoT hash  $\neq$  handoff hash at accept  $\rightarrow$  reject; supervisor re-plans.
- **Idempotency:** Same (`orchestration_run_id`, `target`, `goal`, `state_hash`) does not start a duplicate specialist run if a completed child exists for that key.
- **Timeouts:** Global + per-handoff; cancel leaves audit, no silent retry escalate.
- **No chat handoff** as authority.

### Prohibited delegation paths (examples):

- OBS  $\rightarrow$  mutating tools
- BOPA  $\rightarrow$  submit/pipeline
- Any agent  $\rightarrow$  truth waive



- Specialist → specialist mutate without supervisor admission
- Supervisor → FR-008 repair tools

## 13. Shared-State Model

### No unstructured blackboard.

Authoritative SoTs remain:

- Opportunity
- Workflow checkpoints (recovery only)
- Application package
- TruthReport
- SubmissionAttempt
- PipelineEvents
- AgentRun audit (specialist)
- **New:** OrchestrationRun audit / checkpoints (orchestration layer only)

### Minimum orchestration state

| Field                             | Role                        |
|-----------------------------------|-----------------------------|
| orchestration_run_id              | Parent identity             |
| owner_goal                        | Typed                       |
| opportunity_id or opportunity_ids | Scope                       |
| active_specialist                 | Current child               |
| completed_specialists             | With child run ids + hashes |
| handoff_log_ref                   | Append-only                 |
| global_step_count                 | Budget                      |
| specialist_visit_counts           | Loop control                |
| token_cost_budget                 | Optional; null offline      |
| stop_reason                       | Explicit                    |
| owner_input_required              | Gate                        |
| checkpoint_ref                    | Resume                      |

Agents **re-read** SoT on every observe; conversation memory is never authority.

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## 14. Coordination Model

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| Mode                   | CIC use                                                  |
|------------------------|----------------------------------------------------------|
| Sequential delegation  | <b>Default</b> — OBS then BOPA, or BOPA alone            |
| Conditional delegation | <b>Yes</b> — state class → specialist matrix             |
| Safe parallel          | <b>Read-only only</b> — fan-out OBS across Opportunities |
| Fan-in                 | Aggregate briefs; never merge conflicting mutate results |
| Supervisor loop        | Observe → delegate → await child → re-observe            |
| Finite-state graph     | Preferred mental model for DOS                           |
| Event-driven mesh      | Rejected for FR-016                                      |

**Do not parallelise** writers to the same Opportunity package/truth/pipeline.

Deterministic sequencing remains preferable whenever the next legal mutate action is uniquely determined (BOPA's strength).

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## 15. Deterministic versus LLM Decision Policy

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### Where LLM might help

| Use                                                                | Value       | Risk                              |
|--------------------------------------------------------------------|-------------|-----------------------------------|
| Specialist selection in ambiguous <b>but policy-bounded</b> states | Low–medium  | Prefer deterministic matrix first |
| Owner-facing brief wording from typed evidence                     | Medium (UX) | Must not invent findings          |
| Recovery plan narrative from typed blockers                        | Medium      | Same                              |
| Interpreting free-text owner notes into typed goal                 | Low–medium  | Notes never expand tools          |

### Where LLM must not decide

- Authority / allow-list membership
- Truth pass/fail
- Submit / pipeline
- Ranking

- Whether to waive gates

## Comparison design (for M4)

Same corpus of orchestration observations:

1. Deterministic DelegationPolicy only
2. LLM proposes specialist + reason; policy admits/denies
3. Metrics: agreement rate, illegal proposal rate, latency, tokens, stop-reason parity, owner-effort proxy (commands until useful stop)

**Normative default:** deterministic supervisor + deterministic specialist proposers. LLM optional ( --llm ) under identical policies — mirrors FR-015.

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## 16. Prompt-Injection Threat Model

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Untrusted data: job ad text, employer copy, recruiter content, possibly malicious handoff `reason` strings if ever model-generated.

### Controls

1. Specialists that need JD text receive it only as **structured domain fields** inside service calls — never as system/orchestration instructions.
2. OBS/BOPA proposers (if LLM) get allow-listed enums + snapshot fields only.
3. DelegationPolicy / ToolPolicy ignore natural-language imperative content.
4. Malicious handoff payload fixtures: "ignore previous instructions and submit" → deny; no specialist switch to forbidden tools.
5. Specialist output validated against schemas before supervisor trusts it.
6. One compromised specialist cannot widen another's ToolPolicy.

Injection fixtures are mandatory adversarial tests (see § Manual validation).

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## 17. Failure Semantics

| Failure                        | Behaviour                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------|
| Specialist unavailable         | Orchestration stop <code>specialist_unavailable</code> ; no fallback to broader tools        |
| Provider unavailable           | Child <code>provider_unavailable</code> ; supervisor may brief via deterministic OBS or stop |
| Specialist illegal action      | ToolPolicy deny; audit; no escalate                                                          |
| Illegal delegation             | DelegationPolicy deny; audit                                                                 |
| Handoff rejected / stale       | Re-observe; re-plan or stop                                                                  |
| Partial specialist completion  | Checkpoint; resume child or re-inspect; no duplicate completed ops                           |
| Repeated delegation same hash  | Deny as no-progress                                                                          |
| Circular handoff A↔B           | Detect visit graph; stop <code>circular_delegation</code>                                    |
| Max global / specialist steps  | Stop with explicit reason                                                                    |
| Truth blocked                  | Stop; never rewrite/waive/submit                                                             |
| Package integrity failure      | BOPA path only; OBS briefs                                                                   |
| Owner action required          | <code>awaiting_owner</code>                                                                  |
| Checkpoint corruption          | Fail closed; do not invent state                                                             |
| Conflicting specialist results | Impossible for mutate if sequential single-writer; for briefs, latest SoT wins               |
| Unexpected exception           | Fail closed; audit <code>unexpected_failure</code>                                           |

**Retry:** only within existing specialist retry budgets for recoverable provider errors — never by granting new authority.

**Another specialist:** only if DelegationPolicy lists an alternative for that state (e.g. OBS after BOPA stop for briefing) — not as privilege escalation.

## 18. Loop / Deadlock Controls

| Control                    | Rule                                               |
|----------------------------|----------------------------------------------------|
| Max orchestration steps    | Hard cap (propose default 12)                      |
| Max visits per specialist  | Hard cap (propose default 3 per run)               |
| Repeated handoff detection | Same target+goal+hash → deny                       |
| Circular delegation        | Path cycle detection                               |
| Conflicting requests       | Single active child                                |
| Waiting on each other      | Forbidden — no peer wait; supervisor owns schedule |
| Duplicate work             | Completed-operation keys from child runs           |
| No-progress                | Unchanged observation hash across N steps → stop   |
| Global timeout             | Wall clock budget                                  |
| Cost budget                | When tokens present                                |

**Explicit stop reasons (additive to FR-015):** e.g. `circular_delegation`, `delegation_blocked`, `specialist_unavailable`, `orchestration_max_steps`, `handoff_stale`, `briefing_complete`.

## 19. Audit and Observability Model

### Append-only orchestration events (proposed)

`orchestration_started`, `state_observed`, `specialist_considered`, `specialist_selected`, `delegation_allowed`, `delegation_blocked`, `handoff_created`, `handoff_accepted`, `handoff_rejected`, `specialist_started`, `specialist_completed`, `specialist_stopped`, `domain_refs_cited`, `owner_gate_reached`, `retry_recorded`, `timeout_recorded`, `budget_snapshot`, `orchestration_stop_recorded`.

## Relationships

| Record                               | Role                                 |
|--------------------------------------|--------------------------------------|
| OrchestrationRun                     | Parent projected status + checkpoint |
| Append-only events                   | Reconstructible history              |
| Child AgentRun                       | Existing FR-015 audit                |
| parent_orchestration_run_id on child | Correlation                          |
| Trace / correlation id               | Same across parent/child             |

Projected status is derived; events are authoritative for history (same pattern as pipeline events vs Opportunity status).

### Metrics for FR-017 readiness (collect, don't over-build)

Delegation allow/deny counts; specialist mix; handoff reject/stale rates; steps; tokens/cost; stop-reason histogram; loop prevention hits; owner-gate frequency; deterministic vs LLM agreement.

## 20. Checkpoint and Resume Design

| Concern                           | Design                                                         |
|-----------------------------------|----------------------------------------------------------------|
| Orchestration checkpoint          | Persist orchestration state + last handoff ids                 |
| Specialist checkpoint             | Existing AgentRun checkpoint                                   |
| Owner pause                       | awaiting_owner on parent and/or child                          |
| Safe resume                       | Re-load SoT; recompute observation; validate handoff freshness |
| Stale handoff                     | Reject; new handoff or stop                                    |
| Skip completed specialist work    | Consult completed_specialists + child completed ops            |
| Owner edits during pause          | Snapshot hash change → re-plan (often OBS or BOPA truth path)  |
| Truth/package change during pause | Same                                                           |

Resume never trusts cached LLM memory.

## 21. Owner Workflow

Prefer **extending** `cic agent` over a new `cic orchestrate` namespace unless M1 proves overcrowding.

Conceptual surface:

```
cic agent orchestrate <opportunity_id> --approve
cic agent orchestrate-batch --apply-open --brief-only
cic agent resume <orchestration_run_id> --approve
cic agent show <orchestration_run_id> # detects parent vs child
cic agent history <orchestration_run_id>
```

Exact CLI names are M3 decisions. Owner must see:

- goal
- specialist selected + why (policy reason)
- authority (allow-list summary)
- actions / child run refs
- evidence refs
- stop reason
- next owner action

No role-play chat UI.

## 22. Boundaries: FR-008, FR-013, FR-014, FR-015

| Boundary | FR-016 rule                                                                         |
|----------|-------------------------------------------------------------------------------------|
| FR-008   | Remains acquisition/pre-decision engine; agents diagnose missing artefacts and stop |
| FR-013   | Pipeline read OK; write never from orchestration                                    |
| FR-014   | Truth consume only; fresh PASS + owner still required for external use              |
| FR-015   | BOPA frozen behaviour reused as specialist; do not reopen exit criteria             |
| FR-012   | Out of write path                                                                   |
| FR-009   | Ranking untouched; no prestige agent                                                |

```
FR-008 workflow      FR-015 BOPA      FR-016 DOS
(deterministic graph) (single-agent loop) (delegate + handoff)
      |                |                |
      └──────────┬──────────┘
                  domain services / SoTs
```

Three layers; one set of services; no competing engines.

## 23. Future Job Discovery Compatibility

Preserve:

```
Job Discovery (future)
  → authoritative acquired Opportunity (FR-008/009 path)
  → analysis artefacts
  → DOS / BOPA bounded coordination
  → owner review
  → FR-012 submit (owner)
  → FR-013 pipeline (owner)
```

FR-016 ships only a **placeholder goal/handoff shape** for post-acquisition entry. No Seek/LinkedIn/Indeed connectors, scrapers, or schedulers.

Future ranking (not FR-016) must weigh candidate fit, career value, attainability, competition, effort, and owner outcomes — **not** employer prestige alone.

## 24. Framework Assessment

Evaluated after architecture definition (ADR-003 still applies).

| Framework                                          | Fit with typed contracts   | Det. policy                 | Checkpoint           | Tracing            | Handoffs                  | Testability | Provider coupling | Migration cost |
|----------------------------------------------------|----------------------------|-----------------------------|----------------------|--------------------|---------------------------|-------------|-------------------|----------------|
| Extend custom runtime                              | Excellent                  | Native                      | Already              | Extend             | Design typed              | Excellent   | Low               | Lowest         |
| OpenAI Agents SDK                                  | Medium (map to SDK agents) | Possible but easy to dilute | SDK run state        | Good vendor traces | First-class chat handoffs | Medium      | High OpenAI       | Medium–high    |
| LangGraph                                          | Medium                     | Possible                    | Strong checkpointers | LangSmith          | Supervisor/subagents      | Medium      | Ecosystem         | High           |
| Semantic Kernel / <b>Microsoft Agent Framework</b> | Medium                     | Workflows exist             | Yes in MAF           | Azure-oriented     | Handoff/group patterns    | Medium      | MS ecosystem      | High           |
| CrewAI                                             | Weak for fail-closed CIC   | Weak                        | Weaker fit           | Variable           | Role-play oriented        | Weak        | Variable          | Medium         |



**M0 framework decision: No framework adoption.** Architecture stays portable. Interview study can map CIC DOS ↔ “orchestrate via code” / “agents as tools” without locking the repo.

Revisit only under ADR-003 conditions (many parallel branches, complex multi-interrupt durability, distributed multi-tenant needs, or custom runtime becoming a de-facto framework without benefits).

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## 25. Commercial / Product-Value Assessment

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| Question                            | Answer                                                                                                                                     |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Owner effort removed now?           | Modest: safer brief-first / batch triage; little beyond BOPA for single happy path                                                         |
| What FR-015 cannot do?              | Per-specialist permission isolation; parent orchestration audit; safe read-only batch                                                      |
| Reliability vs overhead?            | Split Prep/Truth would <b>reduce</b> reliability-per-complexity; DOS+OBS+BOPA can be net neutral/positive if OBS stays tiny                |
| Token/latency/debug cost?           | Low if deterministic default; high if LLM coordinator fashion                                                                              |
| Product improvement?                | Small near-term; larger as discovery/batch arrives                                                                                         |
| Employer skill signal?              | Strong <b>if</b> we can explain why we rejected theatre                                                                                    |
| Simpler det. supervisor stronger?   | <b>Yes</b> — that is the recommendation                                                                                                    |
| Evidence multi-agent is worthwhile? | Illegal delegation blocked; circular handoff stopped; OBS cannot mutate; resume no duplicate; batch brief saves owner time; det≈LLM safety |

**Fashion test:** Multi-agent is not valuable because it is trendy. CIC proceeds only with constrained topology or defers.

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## 26. Career / Interview Learning Outcomes

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After FR-016 (constrained), the owner should be able to explain:

1. **When multi-agent is appropriate** — distinct tools/policies/failure domains; not personas for deterministic steps.
2. **vs deterministic workflow** — FR-008 owns known graphs; agents own policy-bounded next-action under uncertainty.
3. **vs single agent** — add agents when allow-lists must diverge or isolation helps audit.
4. **Delegation vs handoff ownership** — CIC keeps supervisor ownership (agents-as-tools analogy); chat handoff ownership is a different pattern.

- 5. **Authority** — policy admission planes; handoffs don't grant privileges.
- 6. **Failure / loops / observability** — explicit stop reasons, visit caps, parent/child traces.
- 7. **Framework literacy without lock-in** — map concepts to LangGraph / Agents SDK / MAF / CrewAI critically.

Study-aid generation is **post-freeze**, not M0.

## 27. Risks and Technical Debt

| Risk                                   | Mitigation                                   |
|----------------------------------------|----------------------------------------------|
| Multi-agent theatre (Prep/Truth split) | Explicit reject in ADR; M1 gate              |
| Second workflow engine                 | Ban FR-008 reimplementations                 |
| Privilege escalation via handoff       | DelegationPolicy + per-specialist ToolPolicy |
| LLM supervisor as authority            | Deterministic default; LLM propose only      |
| Audit sprawl / SoT confusion           | Separate orchestration store; cite refs      |
| Scope creep to discovery/submit        | Milestone non-goals                          |
| Premature framework                    | ADR-003 conditions; M0 rejects               |
| Overbuilding FR-017                    | Metrics hooks only                           |
| Commercial disappointment              | Honest \$25; defer option                    |
| Weakening FR-015 freeze                | Reuse BOPA; don't reopen                     |

## 28. Recommended M0–M4 Milestones

Do not blindly copy FR-015's shape; adapt to constrained multi-agent.

| Milestone | Intent                                | Deliverables                                                                                                                                         | Explicit non-goals                       |
|-----------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>M0</b> | Spike                                 | This document; owner accept/revise/defer                                                                                                             | Code, tests, M1                          |
| <b>M1</b> | Contracts + ADR-008                   | OrchestrationRun , Handoff , DelegationPolicy , specialist registry (BOPA+OBS), parent/child ids; unit tests for policies only                       | Full supervisor loop, CLI, LLM           |
| <b>M2</b> | Supervisor runtime                    | DOS loop; handoff store; budgets; OBS read-only adapters; wire BOPA as child; audit events                                                           | Submit/pipeline/discovery; peer handoffs |
| <b>M3</b> | Owner workflow + optional LLM propose | Extend cic agent for orchestrate/show/resume; optional LLM specialist/brief propose under policy; e2e handoffs                                       | Chat UI; batch ranking; job boards       |
| <b>M4</b> | Eval + OAT + freeze                   | Corpus journeys 1–13; det vs LLM; observability hooks for FR-017; acceptance; docs freeze; study-aid <b>source completeness</b> (not the aid itself) | FR-017 full product; discovery           |

If owner **defers** at M0: record deferral in changelog/roadmap; keep BOPA; no ADR-008.

## 29. Definition of Done

### M0 (this spike)

| Criterion                               | Status                 |
|-----------------------------------------|------------------------|
| Sections 1–31 covered                   | <b>Met</b> (this file) |
| Answers “what problem needs > 1 agent?” | <b>Met</b>             |
| Honest defer option                     | <b>Met</b>             |
| No production code/tests/M1             | <b>Met</b>             |

## FR-016 overall (if accepted)

| Criterion                                              | Expectation                  |
|--------------------------------------------------------|------------------------------|
| Accepted M0                                            | Required                     |
| Accepted ADR-008                                       | Required                     |
| Typed supervisor + specialist contracts                | Required                     |
| Explicit authority model                               | Required                     |
| Typed handoffs                                         | Required                     |
| Deterministic DelegationPolicy                         | Required                     |
| Per-specialist ToolPolicies                            | Required                     |
| Loop/deadlock prevention                               | Required                     |
| Checkpoint/resume                                      | Required                     |
| Append-only audit                                      | Required                     |
| Owner gates                                            | Required                     |
| Prompt-injection protection                            | Required                     |
| Unit / functional / adversarial tests                  | Required                     |
| Corpus evaluation                                      | Required                     |
| Det vs LLM comparison where applicable                 | Required                     |
| Manual owner validation                                | Required                     |
| Acceptance report + docs freeze                        | Required                     |
| No discovery / auto-submit leakage                     | Required                     |
| Study-aid <b>source material</b> complete after freeze | Required                     |
| Visual study aid itself                                | <b>Out of FR-016</b> (later) |

## 30. Study-Aid Source Requirements (Preserve During Implementation)

Do **not** build the study aid in M0–M4 freeze window as a separate product. During implementation, keep artefacts clear enough to later generate:

| Study-aid section         | Source artefacts to preserve                  |
|---------------------------|-----------------------------------------------|
| 60-second recap           | ADR-008 decision + this §1                    |
| Five-minute visual lesson | Topology diagram (§8), authority matrix (§11) |
| Architecture diagrams     | §8, §22, handoff lifecycle                    |
| Key trade-offs            | §7, §25                                       |
| Interview Q&A             | §26 + failure/injection stories from M4       |

Prefer diagrams and tables in eval/ADR docs over chat-only explanations.

## 31. Clear M0 Acceptance Recommendation

### Engineering recommendation

**ACCEPT WITH CONDITIONS** the constrained architecture:

1. **Do** implement **DOS + typed handoffs + DelegationPolicy** on custom contracts.
2. **Do** keep **BOPA** as the mutating specialist (FR-015 frozen).
3. **Do** add **OBS** as a **read-only** specialist with a strictly smaller allow-list.
4. **Do not** split Prep/Truth/Review into theatrical agents.
5. **Do not** adopt an orchestration framework in FR-016.
6. **Do not** give supervisor domain super-user powers.
7. **Do not** implement job discovery, auto-submit, pipeline mutation, or truth waiver.
8. Deterministic routing/proposers are default; LLM optional under policy.
9. If the owner's priority is **only** near-term applications, **DEFER FR-016** instead — record rationale; continue with `cic agent`.

### Proposed ADR (on M1)

**ADR-008 Multi-Agent Orchestration** — DOS + per-specialist ToolPolicy + typed handoffs; reaffirm ADR-003/005/006/007.

### Explicit non-starts until acceptance

- No production implementation
- No production tests
- No M1 coding
- No framework adoption
- No frozen FR behaviour changes

- No study-aid generation

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## Manual Validation Strategy (M1–M4 journeys)

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| ID | Journey                               | Expected                            |
|----|---------------------------------------|-------------------------------------|
| 1  | Happy path prep+truth via DOS→BOPA    | Owner review stop; no submit        |
| 2  | Typed handoff OBS→(supervisor)→BOPA   | Accepted handoff; audit complete    |
| 3  | Illegal delegation (e.g. to submit)   | Blocked; audited                    |
| 4  | Truth blocked                         | Stop; no waiver/rewrite/submit      |
| 5  | Prompt injection in JD / handoff text | No authority change                 |
| 6  | Circular handoff                      | Detected; stop                      |
| 7  | Partial completion resume             | No duplicate prep/truth             |
| 8  | Provider unavailable                  | Fail closed                         |
| 9  | SoT change during pause               | Stale handoff rejected; re-plan     |
| 10 | Owner gate                            | No continue past mandatory approval |
| 11 | Pipeline safety                       | Status unchanged by orchestration   |
| 12 | Cost/step limits                      | Enforced                            |
| 13 | Det vs LLM supervisor propose         | Same corpus; compare safety/cost    |

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## Owner Acceptance

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**Outcome: Accepted with revisions** — 2026-08-06.

Owner accepted the narrow topology (DOS + BOPA + OBS), theatre rejection, learning/substrate primary purpose (no strong near-term commercial claim), typed handoffs, dual policy planes, deterministic default, optional LLM behind policy, BOPA unchanged, and a **mandatory M2 go/no-go** before M3/M4.

M1 delivered: [fr016\\_m1\\_orchestration\\_contracts.md](#); [ADR-008](#).

Historical prompt (retained):

Please reply with one of:

1. **Accept M0 as written** — proceed to M1 contracts + ADR-008 draft (DOS + BOPA + OBS).

2. **Accept with revisions** — list changes (especially: defer OBS; add/remove specialists; force deferral of LLM path; require/forbid CLI namespace).
3. **Defer FR-016** — keep BOPA only until Job Discovery or another genuine permission boundary appears; record in changelog/roadmap.
4. **Reject** — multi-agent not justified for this product; close FR-016 scope with rationale.

No M1 work begins without an explicit choice above.